# ECE 308 Lab 4: Servo Velocity Control

Javier Aldana, Garrison Thompson, Victor Merz
aldana0@purdue.edu | thomp901@purdue.edu | merz4@purdue.edu
Experiment date: March 10, 2026

## 1. Introduction

The aim of this lab is to design a PI controller with a setpoint weight. This is similar to a standard PI controller but is better at avoiding large initial peaks of the control signal. This is achieved by adding an adjustable factor to the setpoint in the error in the proportional element of the controller.

The first step of this design is to derive a closed loop transfer function for the feedback system, then calculating the damping ratio, natural frequency, and control gain. The calculated values are then applied in Simulink and tested. The adjustable factor is then adjusted to one observe its impact, as it then behaves like a standard PI controller. After the adjustable factor is set back to zero, the controller is then tested using hardware.

# 2. Results

## 2.1 Derive Closed Loop Transfer Function $\Omega_l(s)/\Omega_d(s)$

$$Vm(s) = k_p\big(b_s\Omega_d(s) - \Omega_l(s)\big) + k_i(\Omega_d(s) - \Omega_l(s))/s$$

$$\text{Vm(s)} = (k_p b_s + k_i/s)\Omega_d(s) - (k_p + k_i/s)\Omega_l(s)$$

$$\Omega_l = P(s)V_m(s)$$

$$\Omega_l = \frac{b_0}{s+a_0}[(k_p b_s + k_i/s)\Omega_d(s) - (k_p + k_i/s)\Omega_l(s)]$$

$$\Omega_l + \frac{b_0}{s+a_0}[(k_p + k_i/s)\Omega_l(s)] = \frac{b_0}{s+a_0}[(k_p b_s + k_i/s)\Omega_d(s)$$

$$\Omega_l(1 + \frac{b_0}{s+a_0}(k_p + k_i/s) = \frac{b_0}{s+a_0}[(k_p b_s + k_i/s)\Omega_d(s)$$

$$\Omega_l(s)/\Omega_d(s) = \frac{\frac{b_0}{s+a_0}[(k_p b_s + k_i/s)}{(1+\frac{b_0}{s+a_0}(k_p + k_i/s)}$$

2.2 Closed Loop Transfer Function when bsp = 0;

$$\Omega_l(s)/\Omega_d(s) = \frac{k_i b_0}{s^2+s(a_0+k_p b_s)+k_i b_0}$$

$$\Omega_l(s)/\Omega_d(s) = \frac{k_i b_0}{s^2 + s(a_0 + k_p b_s) + k_i b_0}$$

2.3 Damping Ratio and Natural Frequency

PO = 5 = $100e^{\frac{-pi(\varsigma)}{\sqrt{1-\varsigma\text{^}2}}}$

ln(.05) = $\frac{-pi(\varsigma)}{\sqrt{1-\varsigma\text{^}2}}$

ln(.05)^2 = $\frac{pi\text{^}2(\varsigma\text{^}2)}{1-\varsigma\text{^}2}$

$(1-\varsigma\text{^}2)\ln(.05)\text{^}2 = pi\text{^}2(\varsigma\text{^}2)$

$\ln(.05)\text{^}2 = (pi\text{^}2)(\varsigma^2) + (\varsigma^2)\ln(.05)\text{^}2$

$\frac{\ln(.05)\text{^}2}{pi^2+\ln(.05)\text{^}2} = \varsigma^2$

$\varsigma = 0.69$

Tp = 0.2 = $\frac{pi}{w_n\sqrt{1-\varsigma\text{^}2}}$

Wn = $\frac{5pi}{\sqrt{(1-.69\text{^}2)}}$ = 21.7

2.4 Calculating kp and ki

$$w_n^2 = k_i b_0$$

$$w_n = \sqrt{k_i b_0}$$

$$\varsigma = (a_0 + k_p b_0)/(2w_n)$$

$$\varsigma = (a_0 + k_p b_0)/(2\sqrt{k_1 b_0})$$

$$a_0 = 39.37$$

$$b_0 = 60.24$$

$$k_i = w_n^2/b_0$$

$$k_i = \frac{21.7^2}{60.24} = 7.816$$

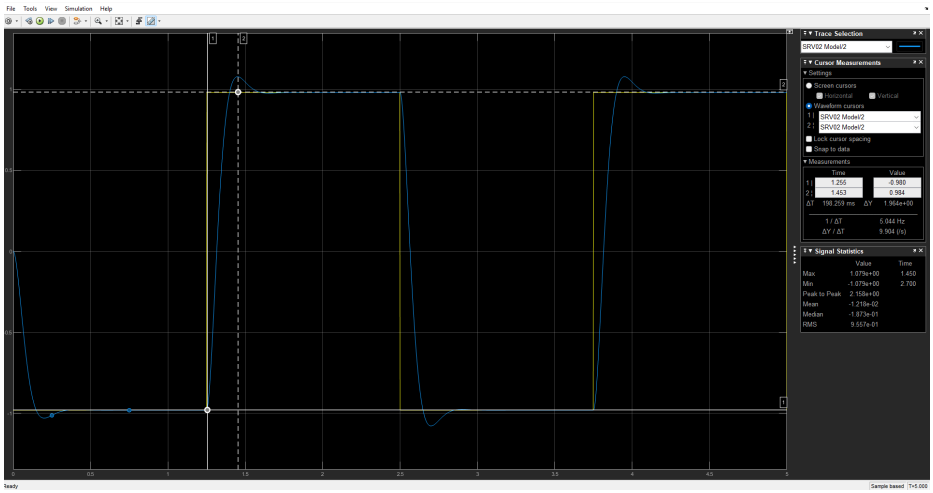


Figure 1: Simulink Step Response using Velocity controller (bsp = 0). X axis: time (sec), Y axis: motor speed (rad/s), yellow trace: reference signal, blue trace: actual output response

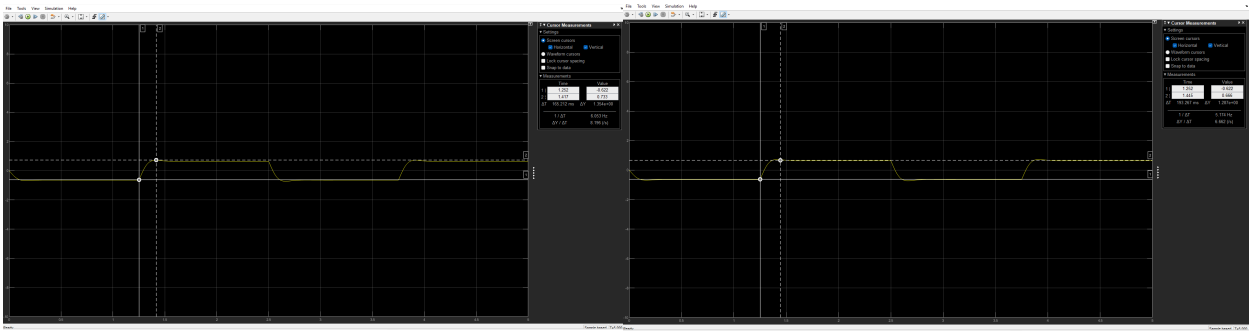


Figure 2: Input Voltage to Velocity controller (bsp = 0). X axis: time (sec), Y axis: voltage (V) yellow trace: input signal

Ss at 0.666V, peak at 0.733V

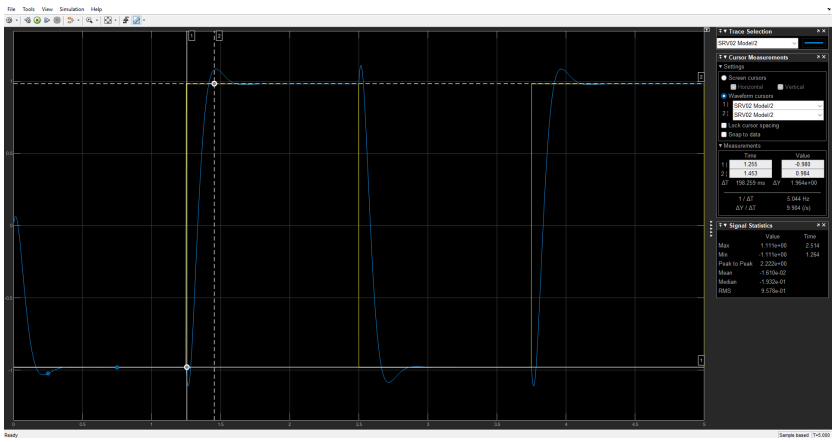


Figure 1: Simulink Step Response using Velocity controller (bsp = 1). X axis: time (sec), Y axis: motor speed (rad/s), yellow trace: reference signal, blue trace: actual output response

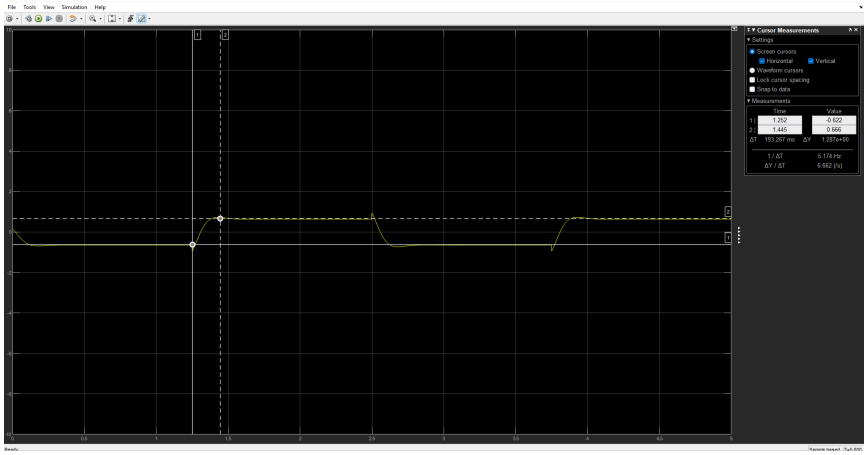


Figure 4: Input Voltage to Velocity controller (bsp = 1). X axis: time (sec), Y axis: voltage (V) yellow trace: input signal

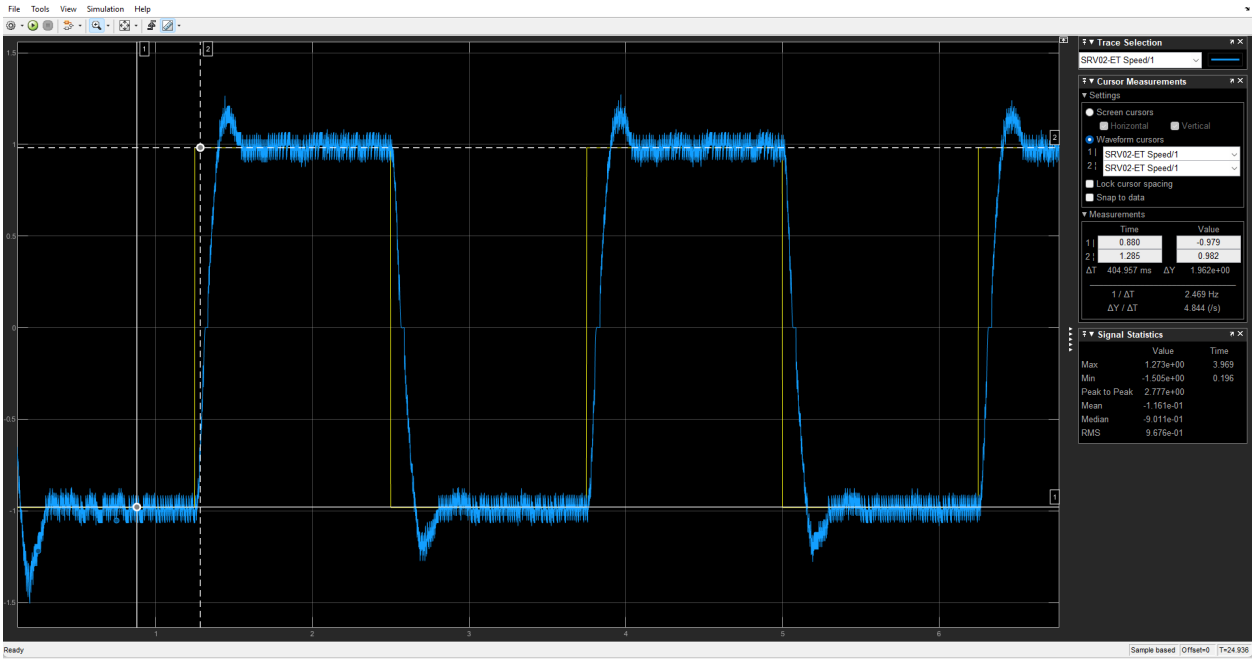



Figure 5: Step Response to Hardware using Velocity controller (bsp = 0). X axis: time (sec), Y axis: motor speed (rad/s), yellow trace: reference signal, blue trace: actual output response

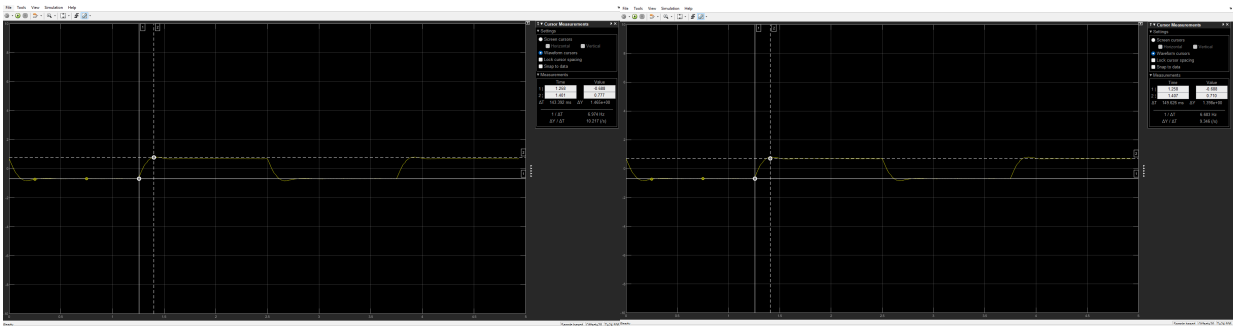


Figure 4: Input Voltage to Velocity controller on Hardware (bsp = 0). X axis: time (sec), Y axis: voltage (V) yellow trace: input signal

Used values:

```
%% Student's Input
%PIV control gains
% Start your work
kp = -0.156;
kv = 0;
ki = 7.817;
% End your work
% ki=39 if you use PIV
```

| | Simulated | Hardware |
|---|---|---|
| Peak Time | 0.2s | 0.22s |
| Percent Overshoot | 10% | 12% |

Table 4.1

# 3. Conclusions

We successfully derived the closed loop transfer function, damping ratio, natural frequency, and control gain of the controller based on the desired parameters. This is shown in Table 4.1. The peak time was very close to the parameters, which is 0.2s, but the overshoot was a bit more, as the desired is 5%, and we had 10%. This could be due to a few factors relating to both the hardware and software, but in general it is around what was intended.

We also observed the impact of the adjustable factor. When it was adjusted to one, there were larger initial peaks in the control signal, but when it was set to zero, the signal was smoother.

In general, the hardware and software yielded similar results, with the biggest difference being that the hardware had a bit more noise. This can be due to things such as friction, which simulations do not account for.